# IMAGING AND DEPLOYMENT LAB

This image boots from the network, and that's just what they do!

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DAVID COGBILL

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## INSTRUCTIONS

**WDS Imaging Project**

### SCENARIO

You are recently hired as a deployment technician and are tasked to setup a simple imaging environment using WDS. Since the company doesn't have existing documentation on this process, they are tasking you to write a KB on the setup of the imaging environment and how to deploy images.
(Document using the normal TF documentation standards)

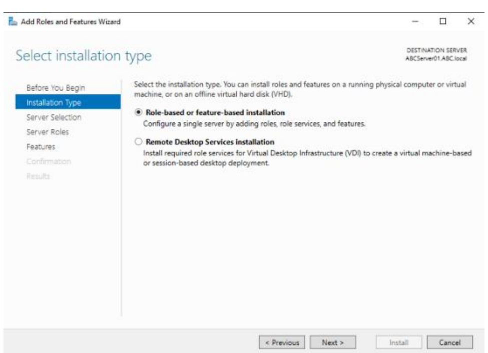
### TO DO

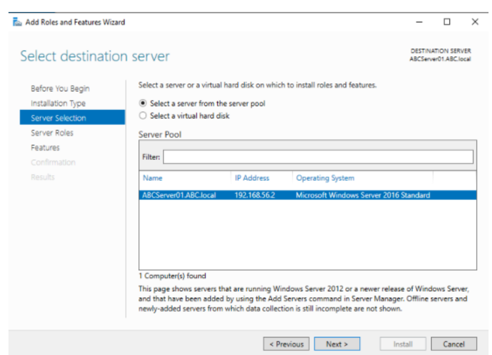
Setting up WDS with a Windows 10 original image steps:
Pre-requisites for the WDS Imaging server: AD DS, DHCP, DNS
WDS role installation and configuration.
Add the corresponding boot.wim and install.wim files from your Windows 10 iso. State what each file does in your documentation.
PXE boot a blank VM and deploy Windows 10 through a network connection with the WDS Server (Make Sure Server and PC are on Host-only Network Adapter Type)
Add any necessary screenshots to support the steps above.

## SETTING UP WINDOWS DEPLOYMENT SERVICES

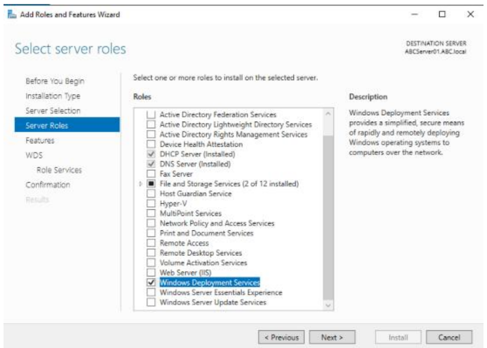
To successfully use WDS (Windows Deployment Service) you must first set up a Windows Server with Active Directory Domain Services, DHCP services, and DNS services. DHCP provides IP addresses and the network boot information to the client. DNS services resolves the name of the WDS server and the name of the Active Directory Domain controller in charge of authentication for the target computer.

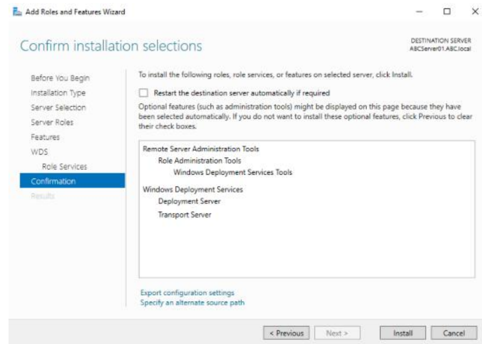
1. Click "Add roles and features" on Server Manager Dashboard window. The Add Roles and Features Wizard window will appear.
2. Ensure "Role-based or feature-based installation" is selected, then click Next.



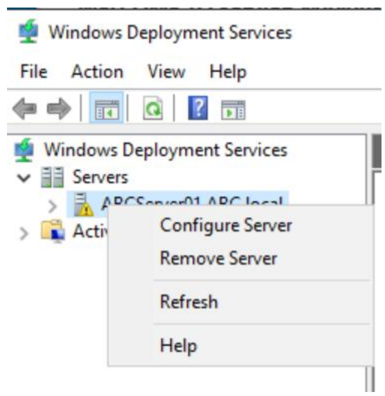




3. Ensure "Select a server from the server pool" and the correct server is selected, then click Next.
4. Select Windows Deployment Services, a new window will appear, then click Add Features.

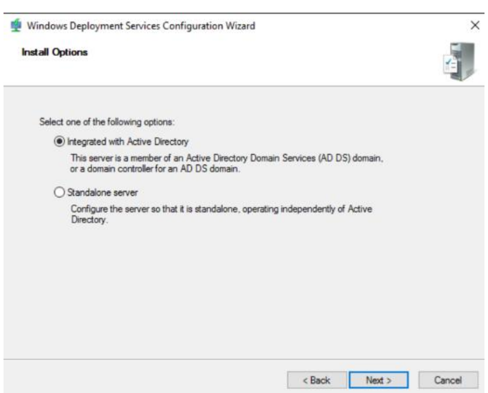


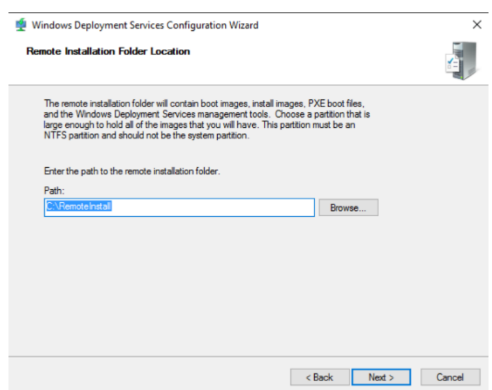




5. Click Next, click Next, click Next, click Next, then click Install.
6. Once the installation is finished, then click Close.
7. Navigate to Windows Deployment Services using the Tools dropdown in the upper right of the Server Manager Dashboard. The Windows Deployment Services window will appear.
8. Click the Server dropdown, right click the server, then click Configure Server. The Windows Deployment Services Configuration window will appear.

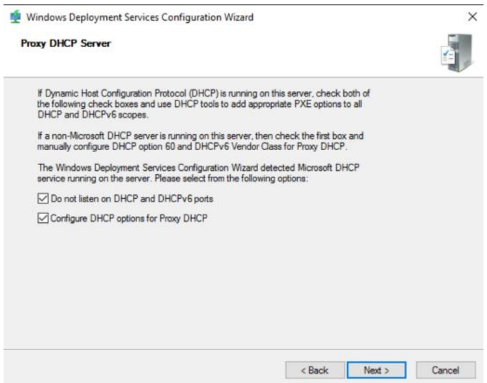


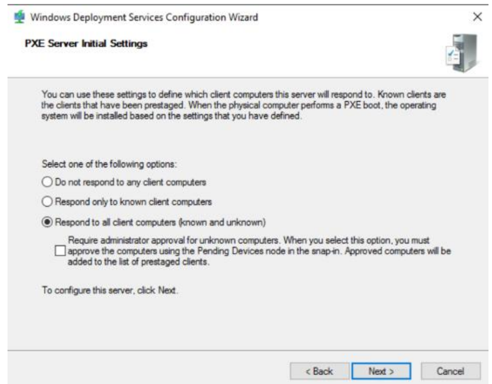

9. Click Next, ensure “Integrated with Active Directory” is selected, then click Next.




10. Here a path can be specified for the remote installation folder. Click Next.
11. A warning will appear when the path is on the windows volume. Click Yes.
12. Leave both selections checked on the Proxy DHCP Server window. Click Next




13. Select “Respond to all client computers (known and unknown)”, click Next, then once the task progress is complete, click Finish.

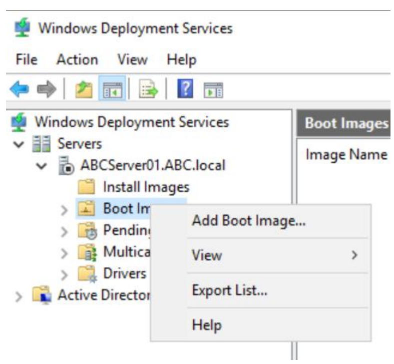
## BOOT.WIM

Boot.wim launches the Windows Pre-Installation Environment, which initiates the setup process. Boot.wim is loaded into a computer’s ram, allows the loading of essential drivers to install hard drives or NICs, and acts as a lighter OS to run the installer initially before handing over to Install.wim.

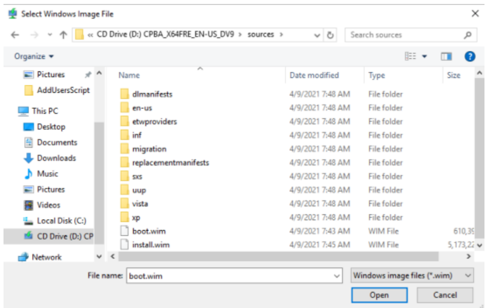
## INSTALL.WIM

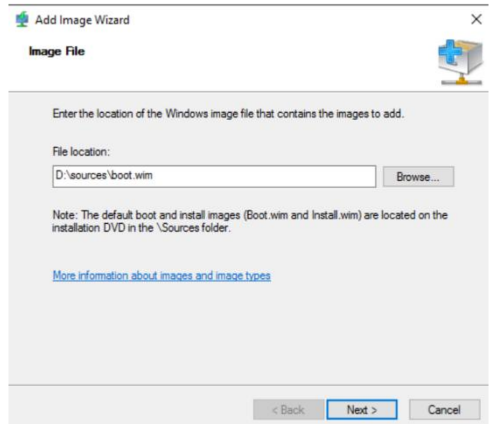
Install.wim is a Windows disk image that can be installed, customized, and deployed, and contains the actual Windows operating system image that is applied to the target computer's hard drive.

14. Right click Boot Image, click Add Boot Image. The Add Image Wizard window will appear.

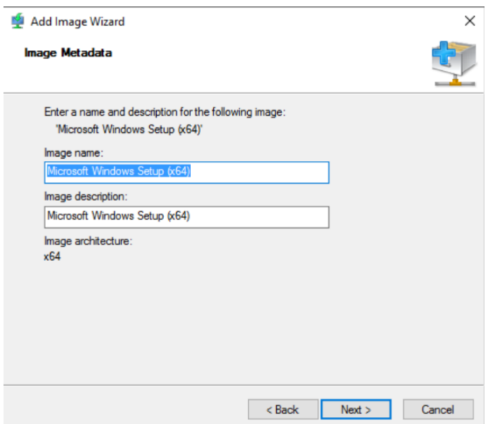


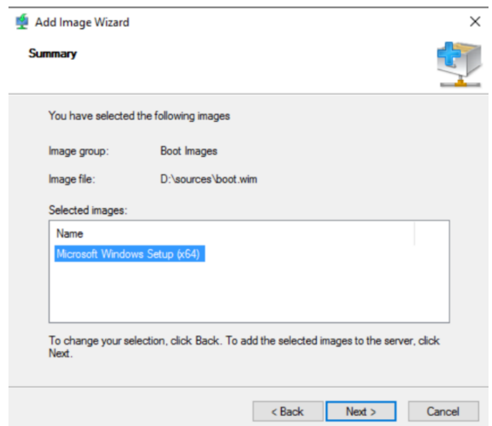

15. Click Browse. Navigate to the “boot.wim” file, inside the sources directory of your Windows 10 ISO. Click Open.



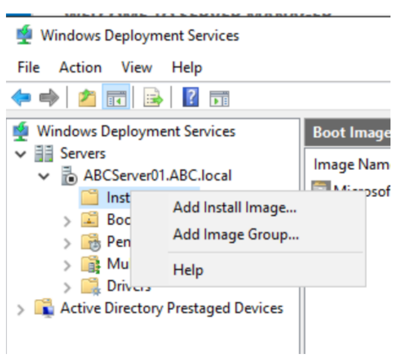




16. Click Next. Here you can enter the name and description of the image, click Next.

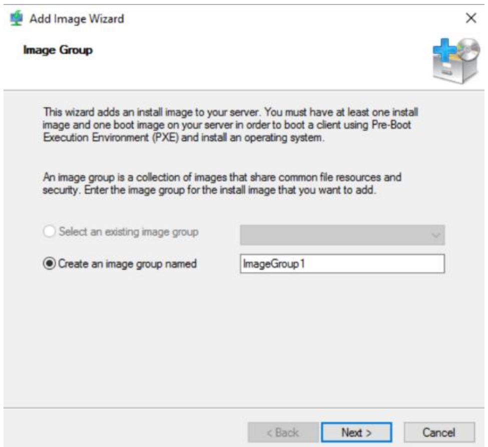


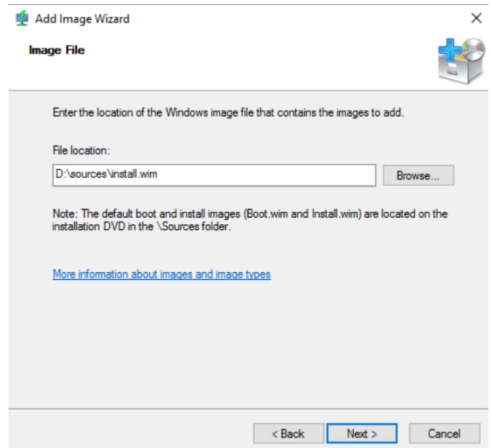




17. Click Next. Once the task progress is complete, click Finish.
18. Right click Install Images, click Add Install Image. The Add Image Wizard window will appear.

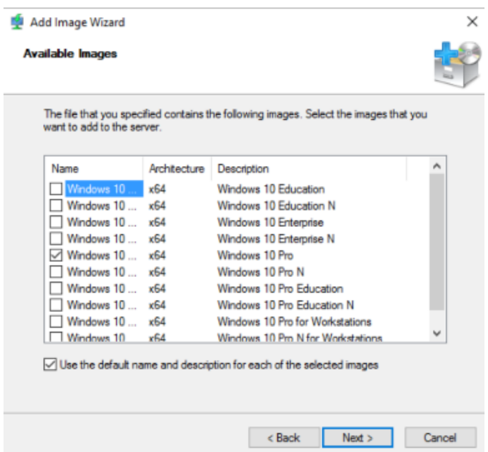


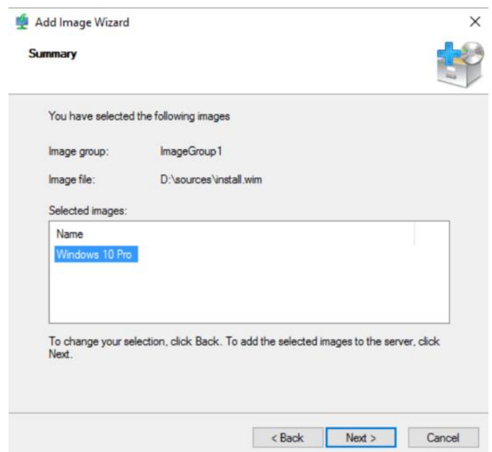

19. Ensure “Create an image group named” is selected and change the name if desired. Click Next.



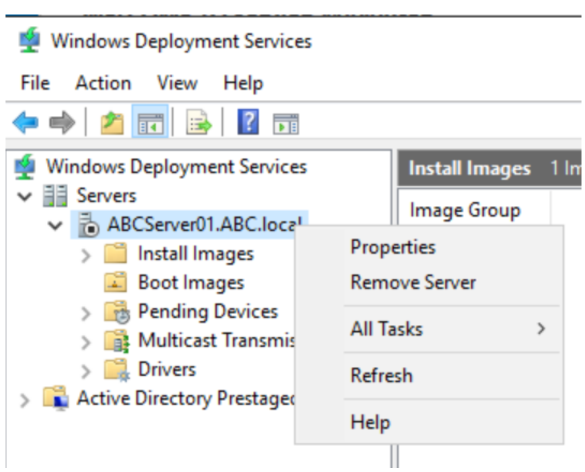




20. Click Browse. Navigate to the "install.wim" file, inside the sources directory of your Windows 10 ISO. Click Open.
21. Here we will be selecting the Windows version we would like to install. Leave only the Windows 10 Pro selected, then click Next.

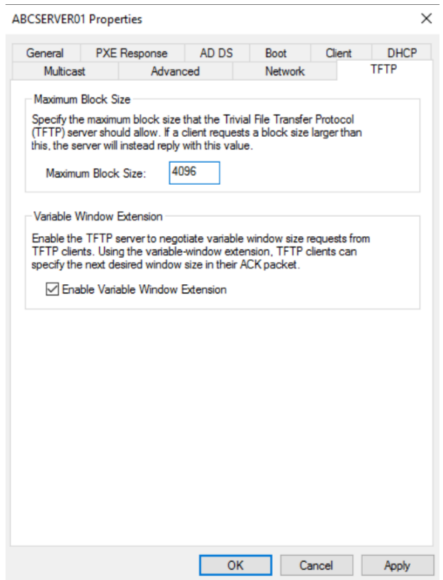


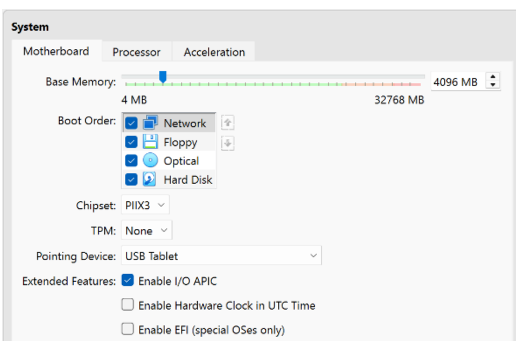

22. Click Next, wait for the task progress to complete, then click Finish.
23. Right click your server, then click properties

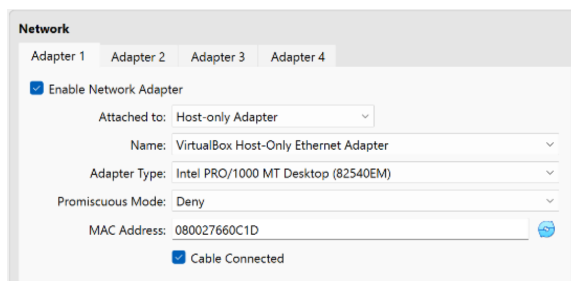


24. Navigate to the TFTP tab, change the Maximum Block Size to 4096 and uncheck "Enable Variable Window Extension", click apply, then click OK.




25. Restart your server, log in, and wait for all services to begin normally.
26. Using the Virtual Box Manager, right-click the designated machine, then click settings. Change the boot order to Network first in the System tab, and the Adapter settings labeled "Attached to:" to "Host-only Adapter" in the Network tab




27. Start the device, be prepared to press F12, click next, then be prompted to log in. Success!